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MAJOR ARTICLE

Prolonged Suppression of Butyrate-Producing Bacteria Is Associated With Acute Gastrointestinal Graft-vs-Host Disease and Transplantation-Related Mortality After Allogeneic Stem Cell Transplantation

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Background. Butyrogenic bacteria play an important role in gut microbiome homeostasis and intestinal epithelial integrity. Previous studies have demonstrated an association between administration of short-chain fatty acids like butyrate and protection from acute graft-vs-host disease (GvHD) after allogeneic stem cell transplantation (ASCT).

Methods. In the current study, we examined the abundance and butyrogenic capacity of butyrate-producing bacteria in 28 healthy donors and 201 patients after ASCT. We prospectively collected serial stool samples and performed polymerase chain reaction analysis of the butyrate-producing bacterial enzyme butyryl–coenzyme A (CoA):acetate CoA-transferase (BCoAT) in fecal nucleic acid extracts.

Results. Our data demonstrate a strong and prolonged suppression of butyrogenic bacteria early in the course of ASCT. In a multivariable analysis, early use of broad-spectrum antibiotics before day 0 (day of transplantation) was identified as an independent factor associated with low BCoAT copy numbers (odds ratio, 0.370 [95% confidence interval, .175–.783]; $P = .009$). Diminished butyrogens correlated with other biomarkers of microbial diversity, such as low 3-indoxylsulfate levels, reduced abundance of Clostridiales and low inverse Simpson and effective Shannon indices (all $P < .001$). Low BCoAT copy numbers at GvHD-onset were correlated with GI-GvHD severity ($P = .002$) and associated with a significantly higher GvHD-associated mortality rate ($P = .04$). Furthermore, low BCoAT copy numbers at day 30 were associated with a significantly higher transplantation-related mortality rate ($P = .02$).

Conclusions. Our results are consistent with the hypothesis that alterations in the microbiome play an important role in GvHD pathogenesis and that microbial parameters such as BCoAT might serve as biomarkers to identify patients at high risk of lethal GI-GvHD.

Keywords. microbiome; butyrate; antibiotic therapy; graft-vs-host disease; allogeneic stem cell transplantation.

Acute graft-vs-host-disease (GvHD) of the gastrointestinal (GI) tract (GI-GvHD) is still one of the most serious complications following hematopoietic allogeneic stem cell transplantation (ASCT) [1]. The impact of intestinal microbiome on GI-GvHD is known to be significant, because studies in mice have shown reduction of GvHD with gut-decontaminating antibiotics and transplantation in germ-free conditions [2, 3]. More recent studies, however, have indicated that clinical use of antibiotics results in microbiota disruption, which is associated with poor outcome after ASCT [4–9]. The intestinal microbiota interact with the host intestinal immune and nonimmune cells and contribute to a broad range of functions in the host, including digestion of nutrients, production of microbial metabolites and maintenance of intestinal homeostasis [10]. Despite a growing understanding of the link between intestinal inflammation and resident gut microbes, the clinical impact of different intestinal microorganisms on acute GvHD is still unclear [11].

Analysis of the gut microbiota has been largely dependent on ex vivo cultivation of bacteria. The development of techniques based on genetic analysis, such as 16S ribosomal DNA sequencing, offers deeper insight into microbiome composition [12]. However, predicting community metabolism solely from taxonomic information is hampered by a limited species-level resolution of 16S ribosomal DNA–based microbiome sequencing, impeding extrapolation of biosynthetic capacities from phenotypically characterized isolates. Beyond that, metabolic information is often missing for noncultivable bacterial species.

In-depth reconstruction of bacterial genomes from stool metagenomic data sets have revealed a high percentage of genomes comprising various phyla that contained butyrate biosynthetic genes. The acetyl–coenzyme A (CoA) pathway was found to be the predominant route for butyrate biosynthesis

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in the intestinal microbiome [13, 14]. In the terminal reaction of this pathway, reduction of butyryl-CoA is accomplished by the butyryl-CoA:acetate CoA-transferase (BCoAT) to form butyrate. On this basis, real-time polymerase chain reaction (PCR)–based quantification of BCoAT gene copy numbers allowed us to examine not only the abundance of butyrate-producing bacteria in healthy donors and patients after ASCT, but also their butyrate-synthesizing capacity.

Butyrogenic bacteria are part of the healthy gut microbiome, and it has been shown that butyrate serves as a major energy source for the gut epithelium; it is a potent histone deacetylase inhibitor and an agonist for peroxisome proliferator-activated receptors, and it is diminished in the intestinal tissue after ASCT [15–17]. Studies have demonstrated an association between administration of butyrate and protection from acute GvHD after ASCT [18, 19]. Butyrate-producing clostridia directly result in the increased presence of regulatory T cells in the gut thereby promoting protection from GVHD development [20–23].

In the current study, we aimed to examine the abundance of butyrogenic bacteria and their butyrate-synthesizing capacity in the course of ASCT. PCR analysis of the functional bacterial enzyme BCoAT was performed on prospectively collected stool samples, and the impact of BCoAT on GvHD and ASCT outcome was investigated.

PATIENTS, MATERIALS, AND METHODS

Patients

A total of 201 adult patients undergoing ASCT between 2008 and 2019 at the University Medical Center of Regensburg were enrolled in this prospective single-center study. The study has been approved by the Ethics Committee of the University Medical Center of Regensburg and was conducted in accordance with the Declaration of Helsinki. After written informed consent, stool and urine specimens were collected at several calendar-driven time points: before conditioning (pretreatment; range, days −9 to − 5; n = 150), on the day of transplantation (day −2 to day 2; n = 37), and on days 7 (days 2–10; n = 135), 14 (11–17; n = 51), 21 (18–24; n = 33), 30 (25–35; n = 110), 60 (40–75; n = 14), and 90 (75–100; n = 10). Stool and urine specimens were also obtained at the onset of GvHD (any organ involvement, n = 69; GI-GvHD, n = 41). Stool samples from 28 healthy donors served as controls.

All donors underwent medical examination and were free of malignant, infectious, or autoimmune diseases; none of them received antibiotic treatment at sample collection. All specimens were stored at −80°C until analysis. Patient and donor characteristics as well as transplantation outcomes are shown in Tables 1 and 2, respectively. Both cohorts were matched for sex (χ^2 = 0.863; P = .35) but overall, donors were younger than patients (P = .03; r = 0.1). GvHD was staged according to the Glucksberg grading system [24].

Table 1. Summary of Patient and Donor Characteristics

Characteristic	Patients or Donors, No. (%)[a]
Age, median (range), y	
Patients	55 (19–69)
Donors	49 (19–68)
Male sex	
Patients	126 (62.7)
Donors	15 (53.6)
Diagnosis	
Acute leukemia	90 (44.8)
Myelodysplastic syndrome	19 (9.5)
Myeloproliferative syndrome	24 (11.9)
Lymphoma	58 (28.9)
Aplastic anemia	5 (2.5)
Other	5 (2.5)
Stage of underlying disease	
Early or intermediate	131 (65.2)
Advanced	70 (34.8)
Donor type	
Sibling	51 (25.4)
Unrelated donor	133 (66.2)
Haploidentical donor	17 (8.5)
Conditioning regimen	
Reduced intensity	175 (87.1)
Standard	26 (12.9)
Gut decontamination	
Ciprofloxacin-metronidazole	10 (5.0)
Rifaximin	191 (95.0)
Antibiotic treatment	
None	15 (7.5)
Before day 0	66 (32.8)
On or after day 0	120 (59.7)
GvHD prophylaxis	
Calcineurin inhibitor + MTX	130 (64.7)
Calcineurin inhibitor + MMF	35 (17.4)
Posttreatment cyclophosphamide	33 (16.4)
Calcineurin inhibitor + bortezomib	3 (1.5)

Abbreviations: GvHD, graft-vs-host-disease; MMF, mycophenolate mofetil; MTX, methotrexate.

[a]Data represent no. (%) of patients or donors unless otherwise specified.

Quantification of Butyryl-CoA:Acetate CoA-Transferase Gene Copy Numbers in the Intestinal Microbiome

All fecal samples were centrifuged, followed by weighting of solid sediments to determine wet weight of fecal contents. Nucleic acids from collected stool samples were isolated as described elsewhere [5]. Subsequently, butyryl-CoA:acetate CoA-transferase gene copy numbers were determined using a degenerated primer set previously published by Louis and Flint [25]. PCR was performed with Roche SYBR Green I Master on the LightCycler 480 II instrument (Roche Diagnostics), applying 40 amplification cycles (95°C, 53°C, and 72°C for 30 seconds each). Melting curved analysis was carried out by continuous florescence measurement during heating from 55°C to 95°C at a rate of 0.5°C per 10 seconds. The butyryl-CoA:acetate CoA-transferase gene of *Roseburia intestinalis* (DSM 14610) cloned

Table 2. Summary of Graft-vs-Host-Disease and Outcome Parameters

Parameter	Patients, No. (%)
Maximum overall grade of acute GvHD	
No acute GvHD	94 (46.8)
Grade I	41 (20.4)
Grade II	46 (22.9)
Grade III	12 (6.0)
Grade IV	8 (4.0)
Maximum stage of GI-GvHD	
No GI-GvHD	147 (73.1)
Stage 1	34 (16.9)
Stage 2	6 (3.0)
Stage 3	8 (4.0)
Stage 4	6 (3.0)
Outcome	
Relapse	47 (23.4)
Survival	140 (69.7)
Death	61 (30.0)
TRM	36 (17.9)
RRM	25 (12.4)

Abbreviations: GI, gastrointestinal; GvHD, graft-vs-host-disease; RRM, relapse-related mortality; TRM, transplantation-related mortality.

into pGEM T Easy vector (Invitrogen) served as a quantification standard.

Analysis of Urinary 3-Indoxyl Sulfate

In 101 patients, urinary 3-indoxylsulfate levels were analyzed between days 0 and 7 by means of reversed-phase liquid chromatography-electrospray ionization-tandem mass spectrometry, as described elsewhere [4]. Some of these data were published elsewhere [4–6, 26].

Analysis of Intestinal Microbiome Diversity

In 116 patients, 16S ribosomal RNA gene analysis of stool specimen collected between days 0 and 7 and quantification of *Clostridium* cluster XIVa species were performed as described elsewhere [5]. The alpha diversity was determined by calculating the inverse Simpson and effective Shannon diversity indices [27, 28]. Some of these data were published elsewhere [4, 5, 26].

Bioinformatics and Data Analysis

Continuous data are presented as median (range) owing to their nonnormal distribution. Accordingly, group comparisons were performed by means of Mann-Whitney *U* test and Wilcoxon sign rank tests, and effect size *r* was calculated according to Rosenthal [29]. Median BCoAT copy numbers at GvHD onset were compared across GI-GvHD stages using the Jonckheere-Terpstra test and post hoc Mann-Whitney *U* test with a Bonferroni correction for multiple comparisons. Correlations were analyzed using Spearman rank correlation coefficients (r_s), with 95% confidence intervals (CIs) calculated according to Bonett and Wright [30]. Absolute and relative frequencies are given for categorical data. Differences in categorial data distribution were analyzed using Pearson χ^2 test. The cumulative incidence of transplantation-related mortality (TRM) was estimated in high- and low-BCoAT groups, considering relapse as a competing risk.

Cox regression and binary logistic regression were used for multivariable assessment of risk factors for TRM and of factors influencing BCoAT copy numbers, respectively. The binary logistic regression model was built with forced entry of predictor variables that had an impact on BCoAT levels at day 0–7 in the previous analysis (use of broad-spectrum antibiotics and gut decontamination). When a patient had BCoAT measured at both day 0 and day 7, the mean value was used for analysis. The Cox regression model was built with forced entry of predictor variables known to be classic risk factors in ASCT (advanced stage of underlying disease, early antibiotic therapy, higher patient age, intensive conditioning, and unrelated donor) and BCoAT copy numbers categorized in either absent or present butyrogenic bacteria at the critical time point for GvHD onset (day 30). All hypotheses were tested on a 2-sided 5% significance level. IBM SPSS Statistics 26 (SPSS) and R (R Core Team 2020) were used for analysis.

RESULTS

Abundance of Butyrogenic Bacteria Correlates With Microbial Diversity

Comparing BCoAT at day 0–7 with isochronal urinary 3-indoxylsulfate levels, an important biomarker of microbial diversity, a positive correlation early after ASCT was found ($P < .001$; $r_s = 0.56$; [95% CI, .38–.70]). Similarly, BCoAT concentrations were correlated with butyrate-producing *Clostridium* cluster XIVa copy numbers ($P < .001$; $r_s = 0.59$ [95% CI .44–.70]), inverse Simpson ($P < .001$; $r_s = 0.43$ [.26–.58]), and effective Shannon indices ($P < .001$; $r_s = 0.52$ [.37–.65]) (Supplementary Figure 1*A*–1*D*).

Strong and Prolonged Suppression of Butyrate-Producing Bacteria Early in ASCT

Analyzing the course of BCoAT during ASCT, there was no difference in copy numbers between healthy donors and pretransplantation samples ($P = .39$; $r = 0.1$) (Figure 1). Although 70 patients (35%) were in an advanced stage of their underlying disease at the time of ASCT, and one might expect a disrupted microbiome at admission owing to intensive pretreatment, there was no difference in the abundance of butyrogens compared with patients with early- or intermediate-stage disease ($P = .43$; $r = 0.1$). Overall a strong and prolonged suppression of BCoAT copy numbers early in ASCT was found, with a significant reduction of BCoAT levels both between pretreatment and day 0 ($P = .01$; $r = 0.5$) and between days 0 and 7 ($P = .003$; $r = 0.6$) (Figure 1). Numbers were equally distributed between patients with and those without GI-GvHD at all time points.

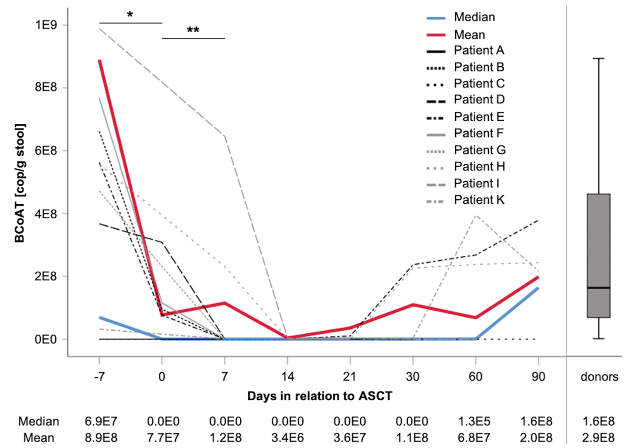

Figure 1. Strong and prolonged suppression of butyrate-producing bacteria early in allogeneic stem cell transplantation (ASCT), with a significant reduction between day −7 and day 0 ($P = .01$; $r = 0.5$), as well as between days 0 and 7 ($P = .003$; $r = 0.6$; Wilcoxon test). Ten representative patient trajectories with median and mean values from the whole cohort are shown. There was no significant difference in copy numbers between healthy donors and pretreatment samples ($P = .39$; $r = 0.1$; Mann-Whitney U test). E notation: 0E0, 0; E2, $\times 10^2$; E4, $\times 10^4$; E6, $\times 10^6$; E8, $\times 10^8$; E10, $\times 10^{10}$. Abbreviation: BCoAT, butyryl–coenzyme A (CoA):acetate CoA-transferase.

Factors Associated With Low BCoAT Copy Numbers: Antibacterial Prophylaxis and Early Treatment With Broad-Spectrum Antibiotics

Searching for factors that might be associated with low BCoAT copy numbers early after ASCT, significantly lower BCoAT copy numbers were found in patients treated with broad-spectrum antibiotics before day 0 during conditioning therapy (n = 48), compared with those without or with delayed antibiotic treatment (starting on or after day 0; n = 98; $P = .005$; $r = 0.2$) (Table 3). A similar effect was observed for different gut decontamination regimens. Our group previously demonstrated that rifaximin preserves intestinal microbiota balance in patients undergoing ASCT [6]. Since 2012, the gut decontamination regimen was changed from ciprofloxacin-metronidazole to rifaximin for all patients receiving ASCT at our center.

When those 2 cohorts were compared, significantly lower BCoAT copy numbers were found at day 0–7 in patients receiving ciprofloxacin-metronidazole (n = 9), compared with those receiving rifaximin (n = 137; $P = .04$; $r = 0.2$). There was no difference in BCoAT copy numbers with respect to intensity of conditioning, stage of underlying disease, and other typical risk factors in ASCT (Table 3). Decontamination and antibiotic treatment were included in a multivariable analysis identifying only antibiotic use before ASCT as an independent parameter for low BCoAT copy numbers at day 0–7 (odds ratio, 0.370 [95% CI, .175–.783]; $P = .009$). In patients in whom no BCoAT copies could be detected (median, 0 copies/g stool), an early start of systemic antibiotic therapy was more frequently observed (Figure 2).

Table 3. Analysis of Factors Influencing Butyryl–Coenzyme A (CoA):Acetate CoA-Transferase Copy Numbers Early After Allogeneic Stem Cell Transplantation

Factor	BCoAT at d 0–7, Median (Range), Copies/g Stool
Antibiotics (before vs on or after d 0)	0 (0–5.4 × 10^8) vs 3.16 × 10^6 (0–2.43 × 10^9)[a]
Gut decontamination (ciprofloxacin-metronidazole vs rifaximin)	0 (0–4.674 × 10^7) vs 0 (0–2.434 × 10^9)[b]
Conditioning (RIC vs standard)	0 (0–2.43 × 10^9) vs 3.16 × 10^6 (0–8.87 4 × 10^8)
Disease stage (early or intermediate vs advanced)	0 (0–2.43 × 10^9) vs 0 (0–7.054 × 10^8)
Patient age (<55 vs ≥55 y)	0 (0–1.02 × 10^9) vs 0 (0–2.43 × 10^9)
Karnofsky index (<90 vs ≥90)	0 (0–7.894 × 10^8) vs 0 (0–2.43 × 10^9)
HCT-CI (<3 vs ≥3)	0 (0–2.43 × 10^9) vs 0 (0–1.94 × 10^9)
Donor type (MRD vs MUD)	0 (0–2.43 × 10^9) vs 0 (0–1.25 × 10^9)

Abbreviations: BCoAT, butyryl–coenzyme A (CoA):acetate CoA-transferase; HCT-CI, hematopoietic cell transplantation-specific comorbidity index; MRD, matched related donor; MUD, matched unrelated donor; RIC, reduced intensity conditioning.

[a] $P = .005$; $r = 0.2$.

[b] $P = .04$; $r = 0.2$.

Protective and Prognostic Effect of BCoAT Copy Numbers at GvHD Onset

Low BCoAT copy numbers at GvHD onset were correlated with GI-GvHD severity (Figure 3). Jonckheere-Terpstra test revealed a significant trend in the data: with decreasing median BCoAT copy numbers, GI-GvHD severity increased ($P = .002$; $r = 0.4$). Patients with gastrointestinal involvement of acute GvHD (n = 43) had lower BCoAT copy numbers at GvHD onset (median, 0 copies/g stool [0–8.31 × 10^8]) than patients with other organ manifestations (n = 26; median, 3.16 × 10^6 copies/g stool [0–7.02]; $P = .006$; $r = 0.3$). All patients in whom severe GI tract involvement developed (n = 10) had no detectable BCoAT copy numbers (median, 0 copies/g stool [0–0]).

Patients with GI-GvHD were next classified into either low- or high-BCoAT groups on BCoAT median copy numbers at GI-GvHD onset (0 copies/g stool [0–8.31 × 10^8]). Competing risk analysis for TRM and relapse is shown in Figure 4A. Because GvHD was the only cause for TRM in this cohort of 41 patients with GI-GvHD, TRM was equivalent to GvHD-associated mortality in this context. Patients with low BCoAT copy numbers displayed a significantly higher GvHD-associated mortality rate than those with high BCoAT concentrations ($P = .04$). Within at the whole study cohort of 201 patients, GI-GvHD–related complications were the primary cause of TRM in 22 of 36 patients (61.1%). Other causes of TRM included infection (30.6 %), hemorrhage (5.6 %), and other organ toxic effects (2.8 %). The mean time to GvHD onset was 32 days after ASCT (95% CI, 26–38 days; range, 9–128 days; median, 26 days), so we combined BCoAT copy numbers in patients with GvHD at GvHD onset with copy numbers at day 30 in patients without GvHD

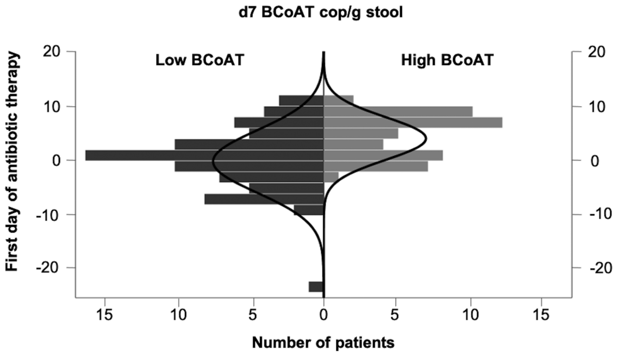

Figure 2. Suppression of butyrogenic bacteria at day 7 after ASCT, in relation to start of antibiotic treatment in 127 patients requiring such treatment. Early systemic antibiotic therapy was more frequently observed in patients with butyryl–coenzyme A (CoA):acetate CoA-transferase (BCoAT) copy numbers below the median.

(controls) and again classified them into either low- or high-BCoAT, groups based on median copy numbers (0 copies/g stool [0–8.31 × 10^8]).

We compared both groups for TRM and found significantly reduced TRM rates in the high BCoAT group ($P = .02$) (Figure 4B). Overall survival rates were lower in patients with low BCoAT copy numbers (0 copies/g stool; n = 32) than in those with high BCoAT concentrations (>0 copies/g stool; n = 9), but this difference was not statistically significant ($P = .08$). Relapse-related mortality rates did not differ between the low- and high-BCoAT groups ($P = .93$). Table 4 shows the results of the Cox proportional hazards analysis for TRM. Low BCoAT copy numbers at day 30 after ASCT were significantly associated with increased risk of TRM ($P = .047$), whereas classic risk factors (eg, age, stage of underlying disease, donor type, conditioning, and early antibiotic therapy) were not. To further illustrate the diagnostic ability of BCoAT levels at day 30 after ASCT to predict TRM, the area under the receiver operating characteristic curve (AUC) was calculated, and the result

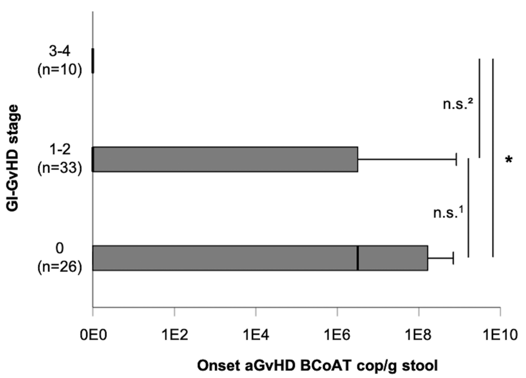

Figure 3. Low butyryl–coenzyme A (CoA):acetate CoA-transferase (BCoAT) copy numbers at onset of acute graft-vs-host disease (GvHD) are correlated with severity of gastrointestinal (GI) GvHD ($P = .002$; $r = 0.3$, Jonckheere-Terpstra test). E notation: 0E0, 0; E5, × 10^5; E6, × 10^6; E7, × 10^7; E8, × 10^8; E9, × 10^9.

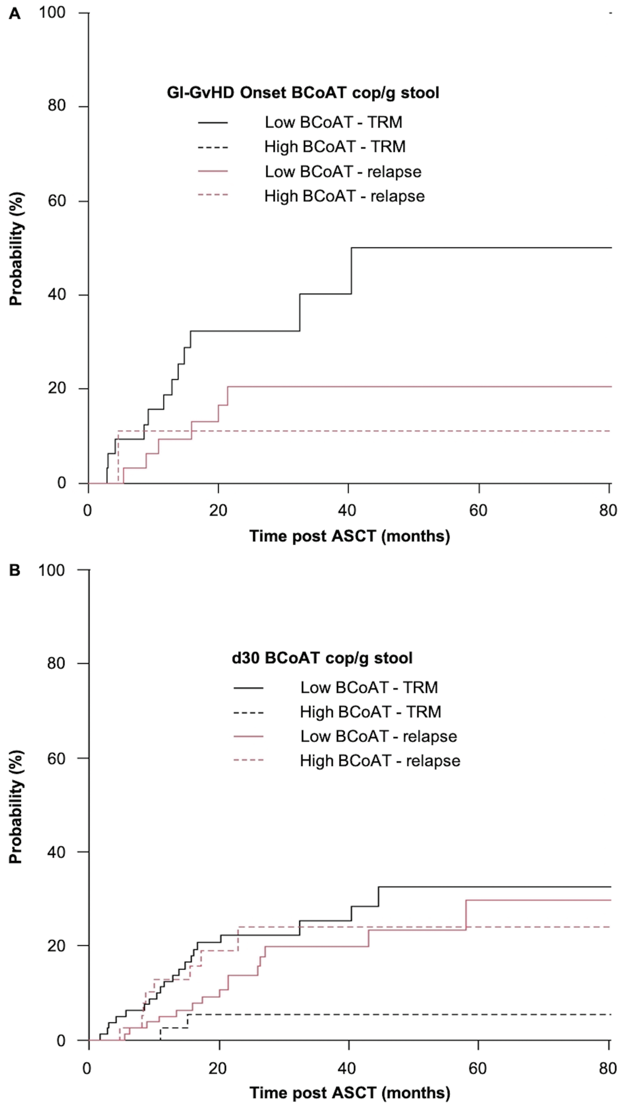

Figure 4. *A,* Competing risk analysis of transplantation-related mortality (TRM) and relapse in relation to median butyryl–coenzyme A (CoA):acetate CoA-transferase (BCoAT) copy numbers at onset of graft-vs-host-disease (GvHD) of the gastrointestinal (GI) tract (GI-GvHD). TRM in patients with low BCoAT copy numbers (0 copies/g stool; n = 32) at GI-GvHD onset was significantly increased compared with that in patients with high BCoAT copy numbers (>0 copies/g stool; n = 9; 50% vs 0%; $P = .04$). Relapse rates did not differ between those groups (21% vs 11%; $P = .23$). *B,* Competing risks analysis of TRM and relapse in relation to median BCoAT copy numbers at day 30 (this includes patients with and without GvHD). TRM in patients with high BCoAT numbers (>0 copies/g stool; n = 39) was significantly reduced compared with patients with low BCoAT copy numbers (n = 81; 33% vs 6%; $P = .02$). Relapse rates did not differ between those groups (30% vs 24%; $P = .72$). Abbreviation: ASCT, allogeneic stem cell transplantation.

was 0.643. This may serve as an encouraging first step toward the integration of microbiota data into predictive GvHD scores.

DISCUSSION

Intestinal microbiota play an important role for the maintenance of intestinal epithelial integrity, immune tolerance and

Table 4. Multivariable Analysis of Factors Influencing Transplantation-Related Mortality

Risk Factor	*P* Value	HR (95% CI)
BCoAT at d 30 + onset (≤0 copies/g stool; n = 81)	.047	4.459 (1.018–19.530)
Stage of underlying disease (advanced; n = 70)	.36	0.609 (.213–1.741)
Early antibiotic therapy (before d 0; n = 66)	.63	0.797 (.318–1.993)
Conditioning (standard; n = 26)	.15	2.108 (.761–5.835)
Donor type (matched unrelated donor; n = 133)	.74	1.094 (.636–1.884)
Patient age (≥55 y; n = 94)	.90	1.056 (.448–2.488)

Abbreviations: BCoAT, butyryl–coenzyme A (CoA):acetate CoA-transferase; CI, confidence interval; HR, hazard ratio.

intestinal homeostasis. Because modern high-throughput sequencing technologies revealed deeper insights into the composition of intestinal microbiota, we have learned to differentiate between commensal bacteria with protective effects, like Lachnospiraceae and Rhuminococcaceae, and bacteria with mainly enteropathogenic influences. Under normal conditions our GI tracts are inhabited by a variety of diverse bacteria [31]. However, during the course of ASCT a loss of intestinal bacterial diversity can be observed ,which is aggravated by the use of systemic antibiotics and particularly present in patients with acute GI-GvHD developing [7].

Several single-center studies [4, 32, 33] and even a multicenter study [34] with almost 9000 fecal samples from >1300 ASCT recipients clearly showed a correlation between low intestinal microbiota diversity and poor clinical outcome. Worse overall survival was due mainly to GI-GvHD–related complications and independent of geographic variations and institutional differences in clinical practice even on different continents [34]. Current research efforts concentrate on the question of how intestinal microbiota contribute to human health or to the development of inflammatory conditions.

Commensal bacteria produce a variety of different microbial metabolites which can regulate immune tolerance. As known so far, these products include short-chain fatty acids (SCFAs) like butyrate, amino acid metabolites indole and its derivatives, and also riboflavin. Intestinal epithelial cells (IECs) are often damaged by the toxicity of the conditioning regimen or GI-GvHD itself, resulting in a damage of the intestinal barrier function. Microbial metabolites can limit the intestinal damage by activation of different pathways, leading to an enhancement of epithelial integrity, improvement of tight junctions, mitigation of inflammation, and regeneration of IECs [35]. Indoles, for example, engage the aryl hydrocarbon receptor and thereby expand innate lymphoid cells and support their production of interleukin 22 that contributes to mucosal immune homeostasis and protection of intestinal tissue [36, 37]. Furthermore, riboflavin metabolites activate mucosal-associated invariant T cells that generate large amounts of interleukin 17 during acute GI-GvHD, leading to maintenance of epithelial barrier function [38]. Therefore, it seems reasonable that a loss of commensal bacteria and their protective metabolites is associated with worse outcome and increased transplantation-related complications after ASCT.

SCFAs are organic acids produced by intestinal microbial fermentation of mainly undigested dietary carbohydrates and comprise acetate, butyrate, propionate and valerate. They are produced mainly in the colon by preferably gram-positive anaerobic bacteria, such as *Faecalibacterium prausnitzii*, belonging to *Clostridium* cluster IV, and *Eubacterium rectale*/*Roseburia* spp., belonging to *Clostridium* cluster XIVa. Butyrate is the major energy source for colonocytes and is therefore involved in the maintenance of colonic mucosal health [18].

There is growing evidence for beneficial effects of SCFAs in the context of ASCT. Several studies directly measured SCFA levels in fecal samples with magnetic resonance spectroscopy or gas chromatography–mass spectrometry. Abundances of butyrogenic bacteria were assessed by taxonomic assignment to species level after 16S ribosomal RNA sequencing [16, 21, 39–41]. In the current study, we combined both approaches by analyzing the bacterial enzyme butyryl-CoA:acetate CoA-transferase (BCoAT), which catalyzes the terminal reaction of the predominant route for butyrate biosynthesis [13, 14]. Real-time PCR–based quantification of BCoAT genes provides a convenient and rapid qualitative assessment of the major butyrate-producing groups present in a given stool sample and provides an excellent source of information on functionally important groups within the colonic microbial community [42].

Our data show a strong and prolonged suppression of butyrate-producing bacteria early in the course of ASCT, with recover starting only about day 60 after transplantation. As expected, suppression of butyrogens was correlated with other markers of diminished microbiome diversity, like low urinary 3-indoxylsulfate levels, reduced abundance of Clostridiales, and low Simpson and Shannon indices in serial fecal samples. Accordingly, early treatment with broad-spectrum antibiotics had significant influence on BCoAT copy numbers. The observed impact of gut decontamination regimen has to be interpreted cautiously owing to the small size of the ciprofloxacin-metronidazole cohort. Patients with acute GI-GvHD had significantly lower copy numbers of butyrogens at GvHD onset than patients without GI involvement, also correlating with an increased risk of GvHD-related mortality. Furthermore, patients with low BCoAT copy numbers at day 30 after ASCT had increased risk for TRM and reduced overall survival. The calculated AUC was not as sensitive as, for example, the AUC for the MAGIC algorithm probability, which has been reported to be 0.81 [43]. Nevertheless this may serve as an encouraging first step toward the integration of microbiota data into predictive GvHD scores, and BCoAT copy numbers will be prospectively analyzed in future studies within the German MAGIC consortium.

Our findings agree with observations of Mathewson et al [19], who found butyrate significantly reduced in murine intestinal tissue early after ASCT, resulting in diminished histone acetylation in IECs. However, restoration of butyrate, either by daily gavage of exogenous butyrate or by butyrate-producing bacteria like Clostridiales, resulted in mitigated GvHD severity and improvement of survival [19]. Stein-Thoeringer et al [44] described enterococcal monodomination in humans and mice early after ASCT, correlating with increased GvHD severity and GvHD-related mortality. In this context enterococcal monodomination was associated with loss of commensal bacteria and loss of fecal butyrate, indicating a key role for this metabolite in causing poor outcomes among monodominated individuals.

These data indicate that loss of microbial metabolites, particularly SCFAs, may play an important role in the pathophysiology of inflammatory disorders like intestinal acute GvHD. However, the molecular mechanism for the effect of microbial-derived metabolites is poorly explored, and direct causality between microbiota and pathogenesis of GvHD is still unclear [35]. In this context G protein–coupled receptors (GPRs) seem to play an important regulatory role. GPR43 is a receptor on the surface of IECs that senses SCFAs like butyrate and propionate in mice. It is critical for protective effects of SCFAs by GPR-mediated extracellular signal-regulated kinase phosphorylation and activation of the nucleotid-binding oligomerization domain (NOD)-like receptor family pyrin domain containing 3 inflammasome in nonhematopoetic target tissues of the host. In this way, GPR43 can protect target tissues from the damage caused by activated allogeneic T cells in mice [45]. The SCFA butyrate acts as an important histone deacetylase inhibitor and therefore displays a preferred energy source of IECs. Reddy et al [15] demonstrated that administration of exogenous histone deacetylase inhibitors significantly reduced secretion of proinflammatory cytokines, allostimulatory responses induced by dendritic cells, and experimental GvHD in murine allogeneic bone marrow transplant models.

Another mechanism for how SCFAs can influence immunological homeostasis in the gut was shown by Furusawa et al [20]. They demonstrated that butyrate induces the differentiation of murine colonic regulatory T cells via T-cell intrinsic epigenetic up-regulation of the Foxp3 gene. Regulatory T cells play a central role in the suppression of inflammatory responses and can ameliorate colitis and intestinal GvHD [20]. Furthermore, butyrate was shown to drive selective macrophage functions toward antimicrobial host defense, again by inhibition of histone deacetylase 3, resulting in maintenance of intestinal homeostasis and microbial diversity [46]. These findings strengthen the assumption that microbial metabolites could be the missing link between intestinal microbiome diversity and intestinal inflammatory conditions.

Our study has some limitations. Patients were not randomized, and we did not take into account all the host variables known to potentially confound gut microbiota studies—for example, geographic variations, general stool quality, body mass index, age, and diet [47]. Further analyses in a larger, multicenter cohort are needed to validate BCoAT as a new potential biomarker to identify patients at high risk for lethal GI-GvHD.

Nevertheless our data show a clear correlation between loss of butyrate-producing bacteria and development of GI-GvHD and increased GvHD-related mortality. Furthermore, the multiple positive effects of SCFAs, such as butyrate, may be exploited clinically for the treatment of GI diseases and potentially even extraintestinal tissue damage [18]. Thus, they are also promising options for the prevention and treatment of severe inflammatory disorders of the GI tract like GI-GvHD, still the most severe and life-threatening complication after ASCT.

Supplementary Data

Supplementary materials are available at *Clinical Infectious Diseases* online. Consisting of data provided by the authors to benefit the reader, the posted materials are not copyedited and are the sole responsibility of the authors, so questions or comments should be addressed to the corresponding author.

Notes

Author Contributions. A. G., E. H., and D. Weber were involved in the conception and design of the study. H. P., M. E., and D. Wolff were responsible for the collection of clinical patient data and collection of specimens. K. D. and G. L. were responsible for analysis of 3-indoxylsulfate levels. A. H. performed polymerase chain reaction analysis. E. M., E. H., and D. Weber collected and analyzed the clinical data and wrote the manuscript. A. H., A. G. S. G., H. P., M. E., W. H., E. H., and D. Weber were involved in the interpretation and discussion of study results. All authors read, revised, and approved the final draft.

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